FuelEU Maritime Compliance Platform

Full-Stack Developer Assignment Submission

March 2026

GitHub Repository
github.com/akansha2026/FuelEU-Maritime

Submission Date
March 22, 2026

Development Time
Approximately 8 hours (AI assisted)

Commit History
28 incremental commits over 2 days

Documentation Files

`README.md`	Project setup, architecture diagram, API documentation, test coverage
`AGENT_WORKFLOW.md`	AI agent prompts, outputs, corrections, and observations
`REFLECTION.md`	Essay on learnings from AI-assisted development

Test Coverage

Layer	Tests	Statements	Branches	Functions
Backend (Node.js)	86	96.13%	91.47%	97.67%
Frontend (React)	141	100%	97.36%	100%
Total	**227**			

Features Implemented

- **Routes Tab** – Filterable table, set regulatory baseline
- **Compare Tab** – Bar chart with 89.34 gCO_2e/MJ target line, compliance status
- **Banking Tab** – Article 20: View CB, bank surplus, apply to deficits
- **Pooling Tab** – Article 21: Select ships, validate sum ≥ 0, greedy allocation
- **Dark/Light Mode** – Smooth transition, localStorage persistence

Technology Stack

Frontend	React 18, TypeScript (strict), TailwindCSS v4, React Query, Recharts
Backend	Node.js, Express 5, TypeScript (strict), Prisma ORM
Database	PostgreSQL 15+
Testing	Vitest, Supertest, React Testing Library
Architecture	Hexagonal (Ports & Adapters / Clean Architecture)

Project Structure

`backend/src/`

- `core/` – Domain, application, ports
- `adapters/` – HTTP, Postgres
- `infrastructure/` – Server, DB
- `shared/` – Constants, errors

`frontend/src/`

- `core/` – Types, ports
- `adapters/ui/` – Components, pages
- `adapters/infrastructure/` – API client
- `shared/` – Constants

AI Assisted Development